SESSION - 1

Graduate Aptitude Test in Engineering

Notations :

1.Options shown in green color and with ✔ icon are correct.

2.Options shown in red color and with ✖ icon are incorrect.

Question Paper Name: ME: MECHANICAL ENGINEERING 31st Jan Shift2
Number of Questions: 65
Total Marks: 100.0

Wrong answer for MCQ will result in negative marks, (-1/3) for 1 mark Questions and (-2/3) for 2 marks Questions.

General Aptitude

Number of Questions: 10
Section Marks: 15.0

Q.1 to Q.5 carry 1 mark each & Q.6 to Q.10 carry 2 marks each.

Question Number : 1 Question Type : MCQ

Choose the appropriate word/phrase, out of the four options given below, to complete the following sentence:

Dhoni, as well as the other team members of Indian team, _____present on the occasion.

(A) were (B) was (C) has (D) have

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 2 Question Type : MCQ

Choose the word most similar in meaning to the given word:

Awkward

(A) Inept (B) Graceful (C) Suitable (D) Dreadful

Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 3 Question Type : MCQ

What is the adverb for the given word below?

Misogynous

(A) Misogynousness (B) Misogynity (C) Misogynously (D) Misogynous

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 4 Question Type : MCQ

An electric bus has onboard instruments that report the total electricity consumed since the start of the trip as well as the total distance covered. During a single day of operation, the bus travels on stretches M, N, O, and P, in that order. The cumulative distances travelled and the corresponding electricity consumption are shown in the Table below:

Stretch	Cumulative distance (km)	Electricity used (kWh)
M	20	12
N	45	25
O	75	45
P	100	57

The stretch where the electricity consumption per km is minimum is

(A) M (B) N (C) O (D) P

Options :

1. ✖ A
2. ✖ B
3. ✖ C
4. ✔ D

Question Number : 5 Question Type : MCQ

Ram and Ramesh appeared in an interview for two vacancies in the same department. The probability of Ram's selection is 1/6 and that of Ramesh is 1/8. What is the probability that only one of them will be selected?

(A) 47/48
(B) 1/4
(C) 13/48
(D) 35/48

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 6 Question Type : MCQ

In the following sentence certain parts are underlined and marked P, Q, and R. One of the parts may contain certain error or may not be acceptable in standard written communication. Select the part containing an error. Choose D as your answer if there is no error.

The student corrected all the errors (P) that the instructor marked (Q) on the answer book (R).

(A) P (B) Q (C) R (D) No Error

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 7 Question Type : MCQ

Given below are two statements followed by two conclusions. Assuming these statements to be true, decide which one logically follows.

Statements:

I. All film stars are playback singers.
II. All film directors are film stars.

Conclusions:

I. All film directors are playback singers.
II. Some film stars are film directors.

(A) Only conclusion I follows.
(B) Only conclusion II follows.
(C) Neither conclusion I nor II follows.
(D) Both conclusions I and II follow.

Options :

1. ✖ A
2. ✖ B
3. ✖ C
4. ✔ D

Question Number : 8 Question Type : NAT

A tiger is 50 leaps of its own behind a deer. The tiger takes 5 leaps per minute to the deer's 4. If the tiger and the deer cover 8 metre and 5metre per leap respectively, what distance in metres will the tiger have to run before it catches the deer?

Correct Answer :

800

Question Number : 9 Question Type : MCQ

If $a^2 + b^2 + c^2 = 1$, then $ab + bc + ac$ lies in the interval

(A) $[1, 2/3]$
(B) $[-1/2, 1]$
(C) $[-1, 1/2]$
(D) $[2, -4]$

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 10 Question Type : MCQ

Lamenting the gradual sidelining of the arts in school curricula, a group of prominent artists wrote to the Chief Minister last year, asking him to allocate more funds to support arts education in schools. However, no such increase has been announced in this year's Budget. The artists expressed their deep anguish at their request not being approved, but many of them remain optimistic about funding in the future.

Which of the statement(s) below is/are logically valid and can be inferred from the above statements?

(i) The artists expected funding for the arts to increase this year.
(ii) The Chief Minister was receptive to the idea of increasing funding for the arts.
(iii) The Chief Minister is a prominent artist.
(iv) Schools are giving less importance to arts education nowadays.

(A) (iii) and (iv) (B) (i) and (iv) (C) (i), (ii) and (iv) (D) (i) and (iii)

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Mechanical Engineering

Number of Questions:	55
Section Marks:	85.0

Q.11 to Q.35 carry 1 mark each & Q.36 to Q.65 carry 2 marks each.

Question Number : 11 Question Type : MCQ

If any two columns of a determinant $P=\begin{vmatrix} 4 & 7 & 8 \\ 3 & 1 & 5 \\ 9 & 6 & 2 \end{vmatrix}$ are interchanged, which one of the following statements regarding the value of the determinant is **CORRECT**?

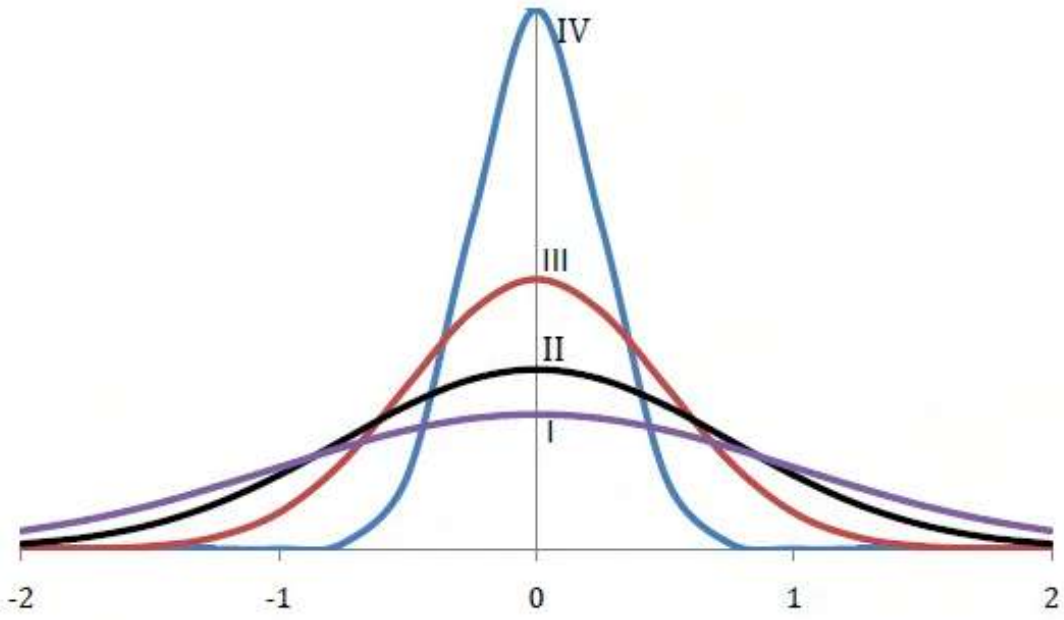
(A) Absolute value remains unchanged but sign will change.
(B) Both absolute value and sign will change.
(C) Absolute value will change but sign will not change.
(D) Both absolute value and sign will remain unchanged.

Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 12 Question Type : MCQ

Among the four normal distributions with probability density functions as shown below, which one has the lowest variance?



(A) I (B) II (C) III (D) IV

Options :

1. ✖ A
2. ✖ B
3. ✖ C
4. ✔ D

Question Number : 13 Question Type : NAT

Simpson's $\frac{1}{3}$ rule is used to integrate the function $f(x) = \frac{3}{5}x^2 + \frac{9}{5}$ between $x = 0$ and $x = 1$ using the least number of equal sub-intervals. The value of the integral is _____

Correct Answer :

2

Question Number : 14 Question Type : MCQ

The value of $\lim_{x \to 0} \frac{1-\cos(x^2)}{2x^4}$ is

(A) 0 (B) $\frac{1}{2}$ (C) $\frac{1}{4}$ (D) undefined

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 15 Question Type : MCQ

Given two complex numbers $z_1 = 5 + (5\sqrt{3})i$ and $z_2 = \frac{2}{\sqrt{3}} + 2i$, the argument of $\frac{z_1}{z_2}$ in degrees is

(A) 0 (B) 30 (C) 60 (D) 90

Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 16 Question Type : MCQ

Consider fully developed flow in a circular pipe with negligible entrance length effects. Assuming the mass flow rate, density and friction factor to be constant, if the length of the pipe is doubled and the diameter is halved, the head loss due to friction will increase by a factor of

(A) 4 (B) 16 (C) 32 (D) 64

Options :

1. ✖ A
2. ✖ B
3. ✖ C
4. ✔ D

Question Number : 17 Question Type : MCQ

The Blasius equation related to boundary layer theory is a

(A) third-order linear partial differential equation
(B) third-order nonlinear partial differential equation
(C) second-order nonlinear ordinary differential equation
(D) third-order nonlinear ordinary differential equation

Options :

1. ✖ A
2. ✖ B
3. ✖ C
4. ✔ D

Question Number : 18 Question Type : MCQ

For flow of viscous fluid over a flat plate, if the fluid temperature is the same as the plate temperature, the thermal boundary layer is

(A) thinner than the velocity boundary layer
(B) thicker than the velocity boundary layer
(C) of the same thickness as the velocity boundary layer
(D) not formed at all

Options :

1. ✖ A
2. ✖ B
3. ✖ C
4. ✔ D

Question Number : 19 Question Type : MCQ

For an ideal gas with constant values of specific heats, for calculation of the specific enthalpy,

(A) it is sufficient to know only the temperature
(B) both temperature and pressure are required to be known
(C) both temperature and volume are required to be known
(D) both temperature and mass are required to be known

Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 20 Question Type : NAT

A Carnot engine (CE-1) works between two temperature reservoirs A and B, where $T_A = 900$ K and $T_B = 500$ K. A second Carnot engine (CE-2) works between temperature reservoirs B and C, where $T_C = 300$ K. In each cycle of CE-1 and CE-2, all the heat rejected by CE-1 to reservoir B is used by CE-2. For one cycle of operation, if the net Q absorbed by CE-1 from reservoir A is 150 MJ, the net heat rejected to reservoir C by CE-2 (in MJ) is _____

Correct Answer :
50

Question Number : 21 Question Type : NAT

Air enters a diesel engine with a density of 1.0 kg/m^3. The compression ratio is 21. At steady state, the air intake is 30×10^{-3} kg/s and the net work output is 15 kW. The mean effective pressure (in kPa) is _____

Correct Answer :
525

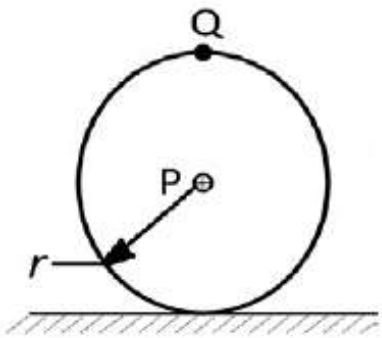
Question Number : 22 Question Type : NAT

A stream of moist air (mass flow rate = 10.1 kg/s) with humidity ratio of 0.01 $\frac{\text{kg}}{\text{kg dry air}}$ mixes with a second stream of superheated water vapour flowing at 0.1 kg/s. Assuming proper and uniform mixing with no condensation, the humidity ratio of the final stream (in $\frac{\text{kg}}{\text{kg dry air}}$) is _____

Correct Answer:
0.02

Question Number : 23 Question Type : NAT

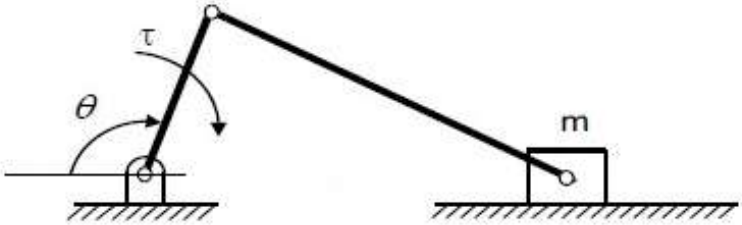
A wheel of radius r rolls without slipping on a horizontal surface shown below. If the velocity of point P is 10 m/s in the horizontal direction, the magnitude of velocity of point Q (in m/s) is _____



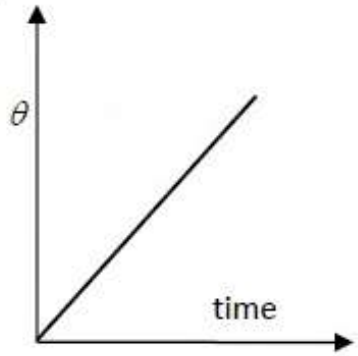
Correct Answer :
20

Question Number : 24 Question Type : MCQ

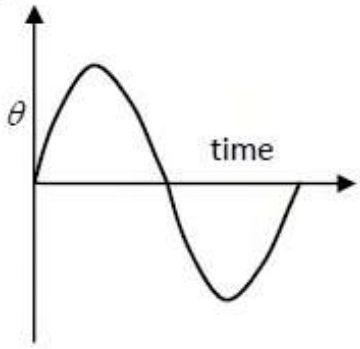
Consider a slider crank mechanism with nonzero masses and inertia. A constant torque τ is applied on the crank as shown in the figure. Which of the following plots best resembles variation of crank angle, θ versus time



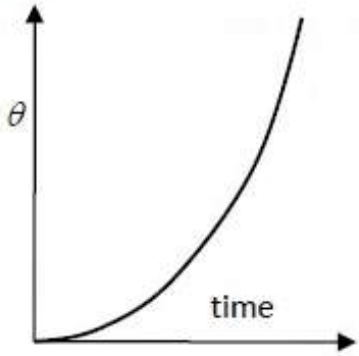
(A)



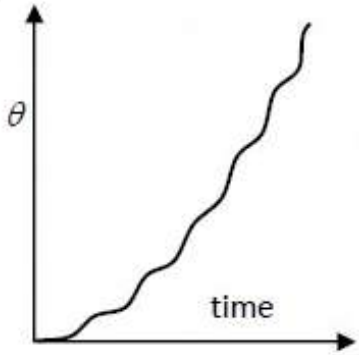
(B)



(C)



(D)

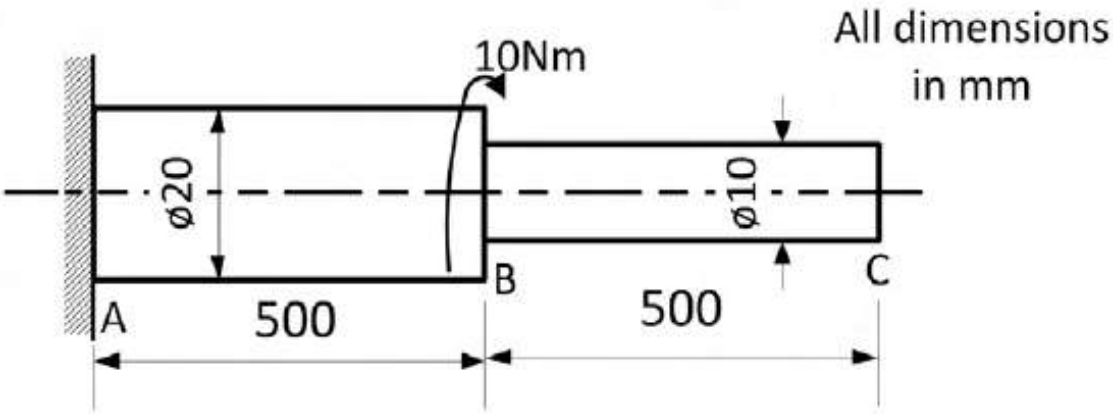


Options :

1. ✖ A
2. ✖ B
3. ✖ C
4. ✔ D

Question Number : 25 Question Type : NAT

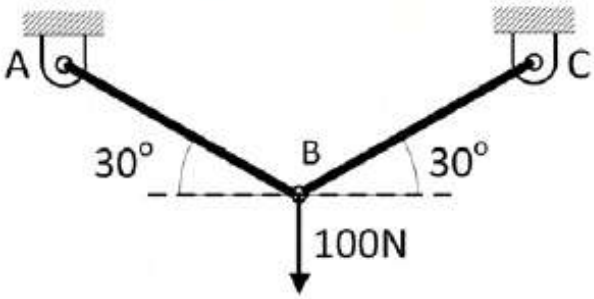
Consider a stepped shaft subjected to a twisting moment applied at B as shown in the figure. Assume shear modulus, G = 77 GPa. The angle of twist at C (in degrees) is _____



Correct Answer:
0.22 to 0.25

Question Number : 26 Question Type : NAT

Two identical trusses support a load of 100 N as shown in the figure. The length of each truss is 1.0 m; cross-sectional area is 200 mm^2; Young's modulus E = 200 GPa. The force in the truss AB (in N) is _____



Correct Answer :
100

Question Number : 27 Question Type : NAT

Consider a steel (Young's modulus E = 200 GPa) column hinged on both sides. Its height is 1.0 m and cross-section is 10 mm × 20 mm. The lowest Euler critical buckling load (in N) is _____

Correct Answer :
3285 to 3295

Question Number : 28 Question Type : NAT

A swimmer can swim 10 km in 2 hours when swimming along the flow of a river. While swimming against the flow, she takes 5 hours for the same distance. Her speed in still water (in km/h) is _____

Correct Answer :

3.5

Question Number : 29 Question Type : MCQ

Which one of the following is the most conservative fatigue failure criterion?

(A) Soderberg
(B) Modified Goodman
(C) ASME Elliptic
(D) Gerber

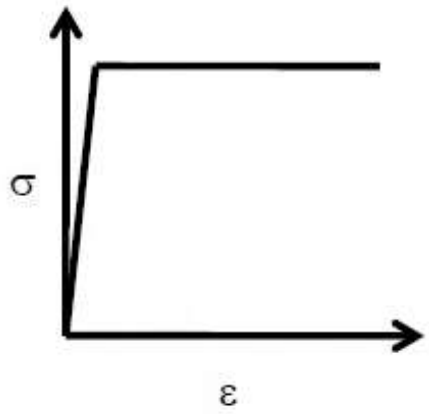
Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

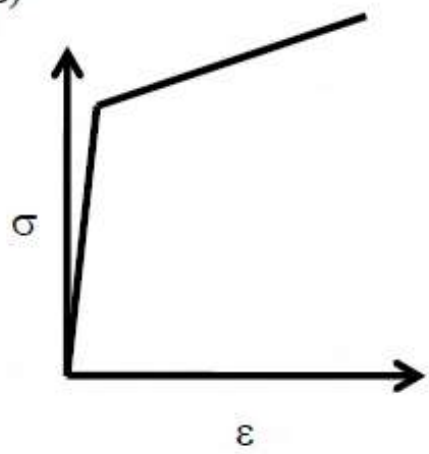
Question Number : 30 Question Type : MCQ

Which one of the following types of stress-strain relationship best describes the behaviour of brittle materials, such as ceramics and thermosetting plastics, (σ = stress and ε = strain)?

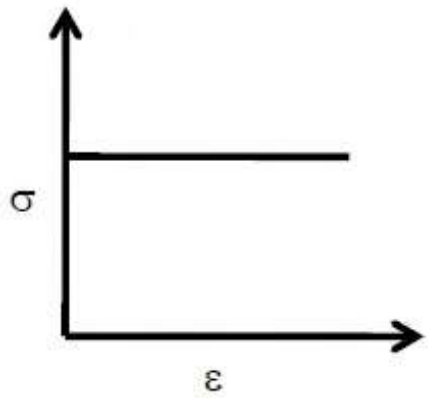
(A)



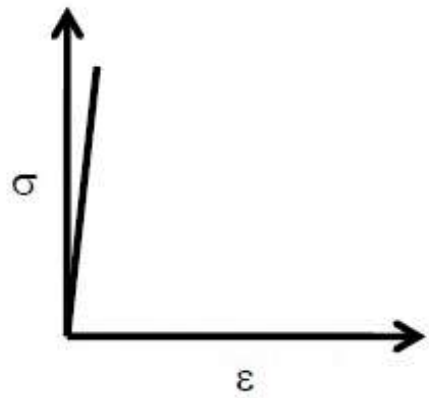
(B)



(C)



(D)



Options :

1. ✖ A
2. ✖ B
3. ✖ C
4. ✔ D

Question Number : 31 Question Type : MCQ

Match the following products with preferred manufacturing processes:

Product		Process	
P	Rails	1	Blow molding
Q	Engine crankshaft	2	Extrusion
R	Aluminium channels	3	Forging
S	PET water bottles	4	Rolling

(A) P-4, Q-3, R-1, S-2
(B) P-4, Q-3, R-2, S-1
(C) P-2, Q-4, R-3, S-1
(D) P-3, Q-4, R-2, S-1

Options :
1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 32 Question Type : MCQ

Holes of diameter $25.0^{+0.040}_{+0.020}$ mm are assembled interchangeably with the pins of diameter $25.0^{+0.005}_{-0.008}$ mm. The minimum clearance in the assembly will be

(A) 0.048 mm
(B) 0.015 mm
(C) 0.005 mm
(D) 0.008 mm

Options :
1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 33 Question Type : NAT

Under certain cutting conditions, doubling the cutting speed reduces the tool life to $\left(\frac{1}{16}\right)^{th}$ of the original. Taylor's tool life index (n) for this tool-workpiece combination will be _____

Correct Answer :
0.25

Question Number : 34 Question Type : MCQ

In a linear arc welding process, the heat input per unit length is inversely proportional to

(A) welding current
(B) welding voltage
(C) welding speed
(D) duty cycle of the power source

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 35 Question Type : MCQ

The function of interpolator in a CNC machine controller is to

(A) control spindle speed
(B) coordinate feed rates of axes
(C) control tool rapid approach speed
(D) perform Miscellaneous (M) functions (tool change, coolant control etc.)

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 36 Question Type : NAT

Consider a spatial curve in three-dimensional space given in parametric form by

$x(t) = \cos t,\ \ y(t) = \sin t,\ z(t) = \frac{2}{\pi}t,\ \ 0 \le t \le \frac{\pi}{2}$.

The length of the curve is _____

Correct Answer :

1.85 to 1.87

Question Number : 37 Question Type : NAT

Consider an ant crawling along the curve $(x-2)^2 + y^2 = 4$, where x and y are in meters. The ant starts at the point (4, 0) and moves counter-clockwise with a speed of 1.57 meters per second. The time taken by the ant to reach the point (2, 2) is (in seconds) _____

Correct Answer :

1.9 to 2.1

Question Number : 38 Question Type : MCQ

Find the solution of $\frac{d^2y}{dx^2} = y$ which passes through the origin and the point $\left(\ln 2, \frac{3}{4}\right)$.

(A) $y = \frac{1}{2}e^x - e^{-x}$

(B) $y = \frac{1}{2}(e^x + e^{-x})$

(C) $y = \frac{1}{2}(e^x - e^{-x})$

(D) $y = \frac{1}{2}e^x + e^{-x}$

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 39 Question Type : MCQ

The probability of obtaining at least two "SIX" in throwing a fair dice 4 times is

(A) 425/432 (B) 19/144 (C) 13/144 (D) 125/432

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

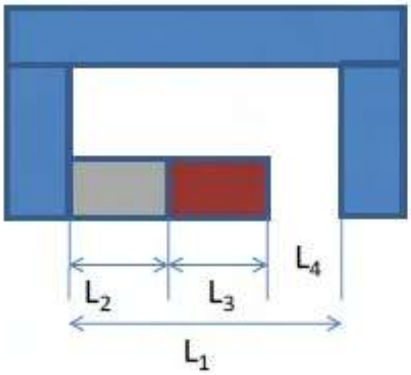
Question Number : 40 Question Type : MCQ

In the assembly shown below, the part dimensions are:

$L_1 = 22.0^{\pm 0.01}$ mm,

$L_2 = L_3 = 10.0^{\pm 0.005}$ mm.

Assuming the normal distribution of part dimensions, the dimension L_4 in mm for assembly condition would be:



(A) $2.0^{\pm 0.008}$

(B) $2.0^{\pm 0.012}$

(C) $2.0^{\pm 0.016}$

(D) $2.0^{\pm 0.020}$

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 41 Question Type : NAT

A DC welding power source has a linear voltage-current (*V-I*) characteristic with open circuit voltage of 80 V and a short circuit current of 300 A. For maximum arc power, the current (in Amperes) should be set as_____

Correct Answer:

149 to 151

Question Number : 42 Question Type : MCQ

A triangular facet in a CAD model has vertices: P1(0,0,0); P2(1,1,0) and P3(1,1,1). The area of the facet is

(A) 0.500
(B) 0.707
(C) 1.414
(D) 1.732

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 43 Question Type : MCQ

Following data refers to the activities of a project, where, node 1 refers to the start and node 5 refers to the end of the project.

Activity	Duration (days)
1-2	2
2-3	1
4-3	3
1-4	3
2-5	3
3-5	2
4-5	4

The critical path (CP) in the network is

(A) 1-2-3-5
(B) 1-4-3-5
(C) 1-2-3-4-5
(D) 1-4-5

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 44 Question Type : NAT

For a canteen, the actual demand for disposable cups was 500 units in January and 600 units in February. The forecast for the month of January was 400 units. The forecast for the month of March considering smoothing coefficient as 0.75 is _____

Correct Answer:

568 to 570

Question Number : 45 Question Type : NAT

An orthogonal turning operation is carried out under the following conditions: rake angle = 5°; spindle rotational speed = 400 rpm; axial feed = 0.4 m/min and radial depth of cut = 5 mm. The chip thickness, t_c, is found to be 3 mm. The shear angle (in degrees) in this turning process is_____

Correct Answer :

18.5 to 19.0

Question Number : 46 Question Type : MCQ

The solidification time of a casting is proportional to $\left(\frac{V}{A}\right)^2$, where V is the volume of the casting and A is the total casting surface area losing heat. Two cubes of same material and size are cast using sand casting process. The top face of one of the cubes is completely insulated. The ratio of the solidification time for the cube with top face insulated to that of the other cube is

(A) $\frac{25}{36}$ (B) $\frac{36}{25}$ (C) 1 (D) $\frac{6}{5}$

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 47 Question Type : NAT

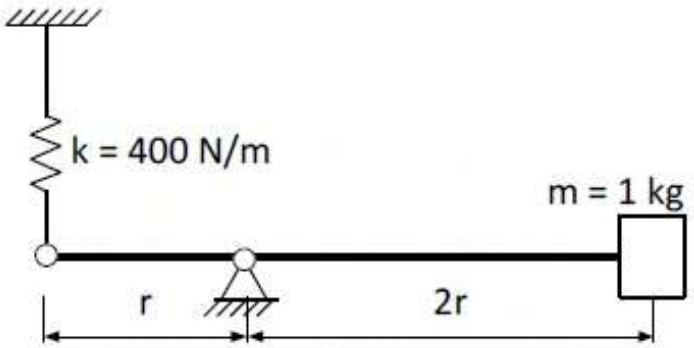
In a slab rolling operation, the maximum thickness reduction (Δh_{max}) is given by $\Delta h_{max} = \mu^2 R$, where R is the radius of the roll and μ is the coefficient of friction between the roll and the sheet. If $\mu = 0.1$, the maximum angle subtended by the deformation zone at the centre of the roll (bite angle in degrees) is _____

Correct Answer :

5.6 to 5.8

Question Number : 48 Question Type : MCQ

Considering massless rigid rod and small oscillations, the natural frequency (in rad/s) of vibration of the system shown in the figure is



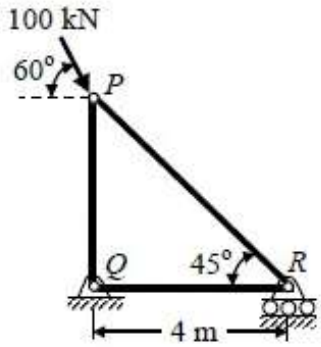
(A) $\sqrt{\frac{400}{1}}$ (B) $\sqrt{\frac{400}{2}}$ (C) $\sqrt{\frac{400}{3}}$ (D) $\sqrt{\frac{400}{4}}$

Options :

1. ✖ A
2. ✖ B
3. ✖ C
4. ✔ D

Question Number : 49 Question Type : MCQ

For the truss shown in figure, the magnitude of the force in member ***PR*** and the support reaction at ***R*** are respectively



(A) 122.47 kN and 50 kN
(B) 70.71 kN and 100 kN
(C) 70.71 kN and 50 kN
(D) 81.65 kN and 100 kN

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 50 Question Type : NAT

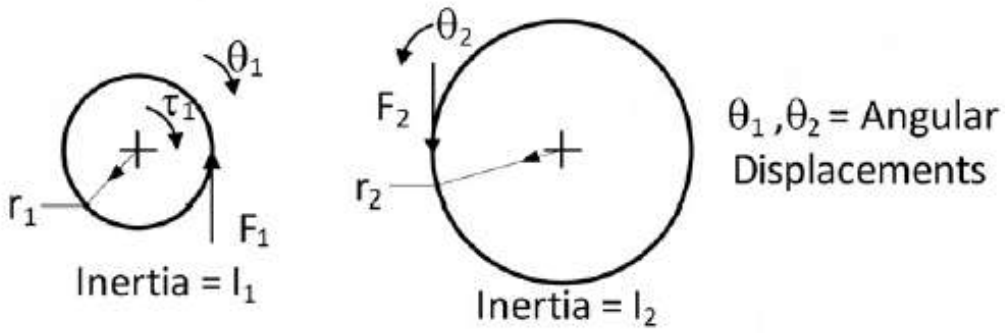
A ball of mass 0.1 kg, initially at rest, is dropped from height of 1 m. Ball hits the ground and bounces off the ground. Upon impact with the ground, the velocity reduces by 20%. The height (in m) to which the ball will rise is _____

Correct Answer:

0.64

Question Number : 51 Question Type : MCQ

A pinion with radius r_1, and inertia I_1 is driving a gear with radius r_2 and inertia I_2. Torque τ_1 is applied on pinion. The following are free body diagrams of pinion and gear showing important forces (F_1 and F_2) of interaction. Which of the following relations hold true?



(A) $F_1 \neq F_2;\ \tau_1 = I_1\ddot{\theta}_1; F_2 = I_2\frac{r_1}{r_2^2}\ddot{\theta}_1$

(B) $F_1 = F_2;\ \tau_1 = \left[I_1 + I_2\left(\frac{r_1}{r_2}\right)^2\right]\ddot{\theta}_1; F_2 = I_2\frac{r_1}{r_2^2}\ddot{\theta}_1$

(C) $F_1 = F_2;\ \tau_1 = I_1\ddot{\theta}_1; F_2 = I_2\frac{1}{r_2}\ddot{\theta}_2$

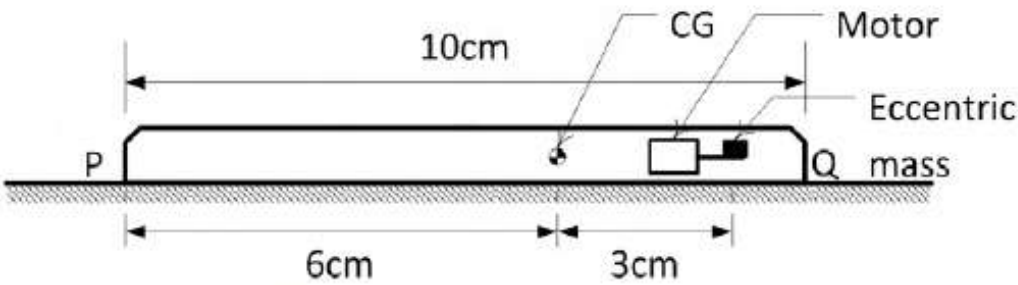
(D) $F_1 \neq F_2;\ \tau_1 = \left[I_1 + I_2\left(\frac{r_1}{r_2}\right)^2\right]\ddot{\theta}_1; F_2 = I_2\frac{1}{r_2}\ddot{\theta}_2$

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 52 Question Type : MCQ

A mobile phone has a small motor with an eccentric mass used for vibrator mode. The location of the eccentric mass on motor with respect to center of gravity (CG) of the mobile and the rest of the dimensions of the mobile phone are shown. The mobile is kept on a flat horizontal surface.



Given in addition that the eccentric mass = 2 grams, eccentricity = 2.19 mm, mass of the mobile = 90 grams, $g = 9.81$ m/s^2. Uniform speed of the motor in RPM for which the mobile will get just lifted off the ground at the end Q is approximately

(A) 3000 (B) 3500 (C) 4000 (D) 4500

Options :

1. ✖ A
2. ✔ B
3. ✖ C

4. ✖ D

Question Number : 53 Question Type : MCQ

A machine element is subjected to the following bi-axial state of stress: $\sigma_x = 80$ MPa; $\sigma_y = 20$ MPa; $\tau_{xy} = 40$ MPa. If the shear strength of the material is 100 MPa, the factor of safety as per Tresca's maximum shear stress theory is

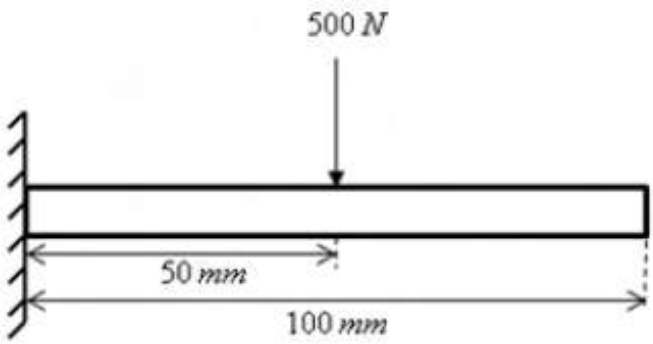
(A) 1.0 (B) 2.0 (C) 2.5 (D) 3.3

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 54 Question Type : NAT

A cantilever beam with flexural rigidity of 200 N.m^2 is loaded as shown in the figure. The deflection (in mm) at the tip of the beam is _____



Correct Answer :

0.24 to 0.28

Question Number : 55 Question Type : NAT

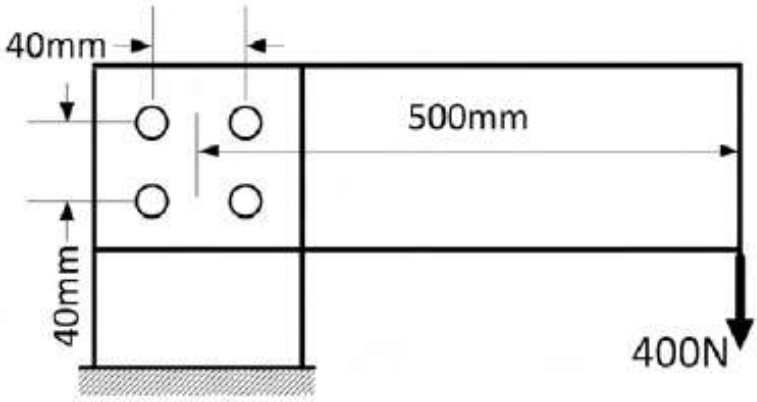
A precision instrument package (m = 1 kg) needs to be mounted on a surface vibrating at 60 Hz. It is desired that only 5% of the base surface vibration amplitude be transmitted to the instrument. Assume that the isolation is designed with its natural frequency significantly lesser than 60 Hz, so that the effect of damping may be ignored. The stiffness (in N/m) of the required mounting pad is _____

Correct Answer :

6750 to 7150

Question Number : 56 Question Type : NAT

A horizontal plate has been joined to a vertical post using four rivets arranged as shown in the figure. The magnitude of the load on the worst loaded rivet (in N) is _____



Correct Answer :

1835 to 1845

Question Number : 57 Question Type : MCQ

For flow through a pipe of radius R, the velocity and temperature distribution are as follows:

$u(r,x) = C_1$, and $T(r,x) = C_2\left[1-\left(\frac{r}{R}\right)^3\right]$, where C_1 and C_2 are constants.

The bulk mean temperature is given by $T_m = \frac{2}{U_m R^2}\int_0^R u(r,x)T(r,x)\,r\mathrm{d}r$, with U_m being the mean velocity of flow. The value of T_m is

(A) $\frac{0.5C_2}{U_m}$ (B) $0.5C_2$ (C) $0.6C_2$ (D) $\frac{0.6C_2}{U_m}$

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 58 Question Type : MCQ

Match the following pairs:

Equation		Physical Interpretation	
P	$\nabla \times \vec{V} = 0$	I	Incompressible continuity equation
Q	$\nabla \cdot \vec{V} = 0$	II	Steady flow
R	$\frac{D\vec{V}}{Dt} = 0$	III	Irrotational flow
S	$\frac{\partial \vec{V}}{\partial t} = 0$	IV	Zero acceleration of fluid particle

(A) P-IV, Q-I, R-II, S-III

(B) P-IV, Q-III, R-I, S-II

(C) P-III, Q-I, R-IV, S-II

(D) P-III, Q-I, R-II, S-IV

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 59 Question Type : NAT

The velocity field of an incompressible flow is given by $\mathbf{V} = (a_1x + a_2y + a_3z)\mathbf{i} + (b_1x + b_2y + b_3z)\mathbf{j} + (c_1x + c_2y + c_3z)\mathbf{k}$, where $a_1 = 2$ and $c_3 = -4$. The value of b_2 is _____

Correct Answer :

1.9 to 2.1

Question Number : 60 Question Type : MCQ

A 10 mm diameter electrical conductor is covered by an insulation of 2 mm thickness. The conductivity of the insulation is 0.08 W/m-K and the convection coefficient at the insulation surface is 10 W/m^2-K. Addition of further insulation of the same material will

(A) increase heat loss continuously
(B) decrease heat loss continuously
(C) increase heat loss to a maximum and then decrease heat loss
(D) decrease heat loss to a minimum and then increase heat loss

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 61 Question Type : NAT

Temperature of nitrogen in a vessel of volume 2 m^3 is 288 K. A U-tube manometer connected to the vessel shows a reading of 70 cm of mercury (level higher in the end open to atmosphere). The universal gas constant is 8314 J/kmol-K, atmospheric pressure is 1.01325 bar, acceleration due to gravity is 9.81 m/s^2 and density of mercury is 13600 kg/m^3. The mass of nitrogen (in kg) in the vessel is _____

Correct Answer :

4.4 to 4.6

Question Number : 62 Question Type : NAT

Air ($\rho = 1.2\ kg/m^3$and kinematic viscosity, $\nu = 2 \times 10^{-5}\ m^2/s$) with a velocity of 2 m/s flows over the top surface of a flat plate of length 2.5 m. If the average value of friction coefficient is $C_f = \frac{1.328}{\sqrt{Re_x}}$, the total drag force (in N) per unit width of the plate is _____

Correct Answer :

0.0158 to 0.0162

Question Number : 63 Question Type : NAT

Water ($\rho = 1000\ kg/m^3$) flows through a venturimeter with inlet diameter 80 mm and throat diameter 40 mm. The inlet and throat gauge pressures are measured to be 400 kPa and 130 kPa respectively. Assuming the venturimeter to be horizontal and neglecting friction, the inlet velocity (in m/s) is _____

Correct Answer :

Question Number : 64 Question Type : NAT

A well insulated rigid container of volume 1 m^3 contains 1.0 kg of an ideal gas [C_p = 1000 J/(kg.K) and C_v = 800 J/(kg.K)] at a pressure of 10^5 Pa. A stirrer is rotated at constant rpm in the container for 1000 rotations and the applied torque is 100 N-m. The final temperature of the gas (in K) is _____

Correct Answer :

1283.4 to 1287.4

Question Number : 65 Question Type : NAT

Steam enters a well insulated turbine and expands isentropically throughout. At an intermediate pressure, 20 percent of the mass is extracted for process heating and the remaining steam expands isentropically to 9 kPa.

Inlet to turbine: P = 14 MPa, T = 560°C, h = 3486 kJ/kg, s = 6.6 kJ/(kg.K)
Intermediate stage: h = 2776 kJ/kg
Exit of turbine: P = 9 kPa, h_f = 174 kJ/kg, h_g = 2574 kJ/kg, s_f = 0.6 kJ/(kg.K), s_g = 8.1 kJ/(kg.K)

If the flow rate of steam entering the turbine is 100 kg/s, then the work output (in MW) is _____

Correct Answer :

123.56 to 127.56

SESSION - 2

Graduate Aptitude Test in Engineering

Notations :

1.Options shown in green color and with ✔ icon are correct.

2.Options shown in red color and with ✖ icon are incorrect.

Question Paper Name: ME : MECHANICAL ENGINEERING 1st Feb Shift1
Number of Questions: 65
Total Marks: 100.0

Wrong answer for MCQ will result in negative marks, (-1/3) for 1 mark Questions and (-2/3) for 2 marks Questions.

General Aptitude

Number of Questions: 10
Section Marks: 15.0

Q.1 to Q.5 carry 1 mark each & Q.6 to Q.10 carry 2 marks each.

Question Number : 1 Question Type : MCQ

Choose the most appropriate word from the options given below to complete the following sentence.

If the athlete had wanted to come first in the race, he _____ several hours every day.

(A) should practise
(B) should have practised
(C) practised
(D) should be practising

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 2 Question Type : MCQ

Choose the most suitable one word substitute for the following expression:

Connotation of a road or way

(A) Pertinacious (B) Viaticum (C) Clandestine (D) Ravenous

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 3 Question Type : MCQ

Choose the correct verb to fill in the blank below:

Let us _____.

(A) introvert (B) alternate (C) atheist (D) altruist

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 4 Question Type : MCQ

Find the missing sequence in the letter series below:
A, CD, GHI, ?, UVWXY

(A) LMN (B) MNO (C) MNOP (D) NOPQ

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 5 Question Type : MCQ

If $x > y > 1$, which of the following must be true?

i. $\ln x > \ln y$ ii. $e^x > e^y$ iii. $y^x > x^y$ iv. $\cos x > \cos y$

(A) (i) and (ii) (B) (i) and (iii)
(C) (iii) and (iv) (D) (ii) and (iv)

Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 6 Question Type : MCQ

Ram and Shyam shared a secret and promised to each other that it would remain between them. Ram expressed himself in one of the following ways as given in the choices below. Identify the correct way as per standard English.

(A) It would remain between you and me.
(B) It would remain between I and you.
(C) It would remain between you and I.
(D) It would remain with me.

Options :

1. ✔ A

2. ✖ B

3. ✖ C

4. ✖ D

Question Number : 7 Question Type : MCQ

In the following question, the first and the last sentence of the passage are in order and numbered 1 and 6. The rest of the passage is split into 4 parts and numbered as 2, 3, 4, and 5. These 4 parts are not arranged in proper order. Read the sentences and arrange them in a logical sequence to make a passage and choose the correct sequence from the given options.

1. On Diwali, the family rises early in the morning.
2. The whole family, including the young and the old enjoy doing this.
3. Children let off fireworks later in the night with their friends.
4. At sunset, the lamps are lit and the family performs various rituals.
5. Father, mother, and children visit relatives and exchange gifts and sweets.
6. Houses look so pretty with lighted lamps all around.

(A) 2,5,3,4 (B) 5,2,4,3 (C) 3,5,4,2 (D) 4,5,2,3

Options :

1. ✖ A

2. ✔ B

3. ✖ C

4. ✖ D

Question Number : 8 Question Type : NAT

From a circular sheet of paper of radius 30 cm, a sector of 10% area is removed. If the remaining part is used to make a conical surface, then the ratio of the radius and height of the cone is _____.

Correct Answer :

1.9 to 2.2

Question Number : 9 Question Type : MCQ

$\log \tan 1^\circ + \log \tan 2^\circ + \ldots\ldots + \log \tan 89^\circ$ is....

(A) 1 (B) $1/\sqrt{2}$ (C) 0 (D) -1

Options :

1. ✖ A

2. ✖ B

3. ✔ C

4. ✖ D

Question Number : 10 Question Type : MCQ

Ms. X will be in Bagdogra from 01/05/2014 to 20/05/2014 and from 22/05/2014 to 31/05/2014. On the morning of 21/05/2014, she will reach Kochi via Mumbai.

Which one of the statements below is logically valid and can be inferred from the above sentences?

(A) Ms. X will be in Kochi for one day, only in May.
(B) Ms. X will be in Kochi for only one day in May.
(C) Ms. X will be only in Kochi for one day in May.
(D) Only Ms. X will be in Kochi for one day in May.

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Mechanical Engineering

Number of Questions:	55
Section Marks:	85.0

Q.11 to Q.35 carry 1 mark each & Q.36 to Q.65 carry 2 marks each.

Question Number : 11 Question Type : MCQ

At least one eigenvalue of a singular matrix is

(A) positive (B) zero (C) negative (D) imaginary

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 12 Question Type : MCQ

At $x = 0$, the function $f(x) = |x|$ has

(A) a minimum
(B) a maximum
(C) a point of inflexion
(D) neither a maximum nor minimum

Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 13 Question Type : MCQ

Curl of vector $\mathbf{V}(x, y, z) = 2x^2\,\mathbf{i} + 3z^2\,\mathbf{j} + y^3\,\mathbf{k}$ at $x = y = z = 1$ is

(A) $-3\mathbf{i}$ (B) $3\mathbf{i}$ (C) $3\mathbf{i} - 4\mathbf{j}$ (D) $3\mathbf{i} - 6\mathbf{k}$

Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 14 Question Type : MCQ

The Laplace transform of e^{i5t} where $i = \sqrt{-1}$, is

(A) $\dfrac{s-5i}{s^2-25}$ (B) $\dfrac{s+5i}{s^2+25}$ (C) $\dfrac{s+5i}{s^2-25}$ (D) $\dfrac{s-5i}{s^2+25}$

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 15 Question Type : NAT

Three vendors were asked to supply a very high precision component. The respective probabilities of their meeting the strict design specifications are 0.8, 0.7 and 0.5. Each vendor supplies one component. The probability that out of total three components supplied by the vendors, at least one will meet the design specification is _____

Correct answer :

0.96 to 0.98

Question Number : 16 Question Type : NAT

A small ball of mass 1 kg moving with a velocity of 12 m/s undergoes a direct central impact with a stationary ball of mass 2 kg. The impact is perfectly elastic. The speed (in m/s) of 2 kg mass ball after the impact will be _____

Correct Answer :

7.8 to 8.2

Question Number : 17 Question Type : NAT

A rod is subjected to a uni-axial load within linear elastic limit. When the change in the stress is 200 MPa, the change in the strain is 0.001. If the Poisson's ratio of the rod is 0.3, the modulus of rigidity (in GPa) is _____

Correct Answer :

76 to 78

Question Number : 18 Question Type : MCQ

A gas is stored in a cylindrical tank of inner radius 7 m and wall thickness 50 mm. The gage pressure of the gas is 2 MPa. The maximum shear stress (in MPa) in the wall is

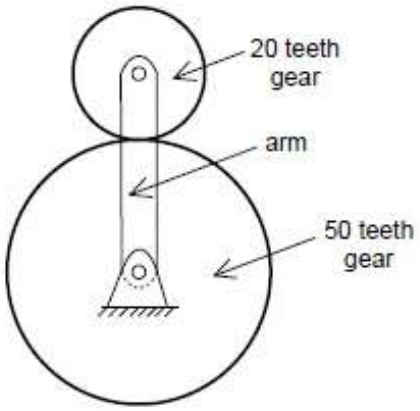
(A) 35 (B) 70 (C) 140 (D) 280

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 19 Question Type : MCQ

The number of degrees of freedom of the planetary gear train shown in the figure is



(A) 0 (B) 1 (C) 2 (D) 3

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 20 Question Type : NAT

In a spring-mass system, the mass is m and the spring constant is k. The critical damping coefficient of the system is 0.1 kg/s. In another spring-mass system, the mass is $2m$ and the spring constant is $8k$. The critical damping coefficient (in kg/s) of this system is _____

Correct Answer :
0.38 to 0.42

Question Number : 21 Question Type : NAT

The uniaxial yield stress of a material is 300 MPa. According to von Mises criterion, the shear yield stress (in MPa) of the material is _____

Correct Answer :
171 to 175

Question Number : 22 Question Type : MCQ

If the fluid velocity for a potential flow is given by $\mathbf{V}(x,y) = u(x,y)\mathbf{i} + v(x,y)\mathbf{j}$ with usual notations, then the slope of the potential line at (x,y) is

(A) $\frac{v}{u}$ (B) $-\frac{u}{v}$ (C) $\frac{v^2}{u^2}$ (D) $\frac{u}{v}$

Options :
1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 23 Question Type : MCQ

Which of the following statements regarding a Rankine cycle with reheating are **TRUE**?
(i) increase in average temperature of heat addition
(ii) reduction in thermal efficiency
(iii) drier steam at the turbine exit

(A) only (i) and (ii) are correct
(B) only (ii) and (iii) are correct
(C) only (i) and (iii) are correct
(D) (i), (ii) and (iii) are correct

Options :
1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 24 Question Type : MCQ

Within a boundary layer for a steady incompressible flow, the Bernoulli equation

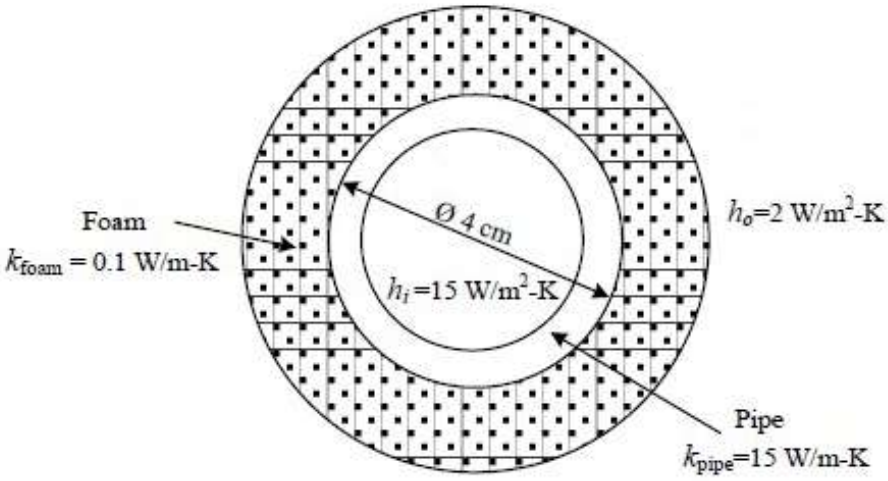
(A) holds because the flow is steady
(B) holds because the flow is incompressible
(C) holds because the flow is transitional
(D) does not hold because the flow is frictional

Options :

1. ✖ A
2. ✖ B
3. ✖ C
4. ✔ D

Question Number : 25 Question Type : NAT

If a foam insulation is added to a 4 cm outer diameter pipe as shown in the figure, the critical radius of insulation (in cm) is _____



Correct Answer :

4.9 to 5.1

Question Number : 26 Question Type : MCQ

In the laminar flow of air (Pr = 0.7) over a heated plate, if δ and δ_T denote, respectively, the hydrodynamic and thermal boundary layer thicknesses, then

(A) $\delta = \delta_T$
(B) $\delta > \delta_T$
(C) $\delta < \delta_T$
(D) $\delta = 0$ but $\delta_T \neq 0$

Options :

1. ✖ A
2. ✖ B
3. ✔ C

4. ✖ D

Question Number : 27 Question Type : NAT

The COP of a Carnot heat pump operating between 6 °C and 37 °C is _____

Correct Answer :

9.8 to 10.2

Question Number : 28 Question Type : MCQ

The Van der Waals equation of state is $\left(p+\frac{a}{v^2}\right)(v-b)=RT$, where p is pressure, v is specific volume, T is temperature and R is characteristic gas constant. The SI unit of a is

(A) J/kg-K (B) m^3/kg (C) m^5/kg-s^2 (D) Pa/kg

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 29 Question Type : NAT

A rope-brake dynamometer attached to the crank shaft of an I.C. engine measures a brake power of 10 kW when the speed of rotation of the shaft is 400 rad/s. The shaft torque (in N-m) sensed by the dynamometer is _____

Correct Answer :

24 to 26

Question Number : 30 Question Type : SA

The atomic packing factor for a material with body centered cubic structure is _____

Correct Answer :

0.66 to 0.69

Question Number : 31 Question Type : MCQ

The primary mechanism of material removal in electrochemical machining (ECM) is

(A) chemical corrosion
(B) etching
(C) ionic dissolution
(D) spark erosion

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 32 Question Type : MCQ

Which one of the following statements is **TRUE**?

(A) The 'GO' gage controls the upper limit of a hole
(B) The 'NO GO' gage controls the lower limit of a shaft
(C) The 'GO' gage controls the lower limit of a hole
(D) The 'NO GO' gage controls the lower limit of a hole

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 33 Question Type : MCQ

During the development of a product, an entirely new process plan is made based on design logic, examination of geometry and tolerance information. This type of process planning is known as

(A) retrieval
(B) generative
(C) variant
(D) group technology based

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 34 Question Type : NAT

Annual demand of a product is 50000 units and the ordering cost is Rs. 7000 per order. Considering the basic economic order quantity model, the economic order quantity is 10000 units. When the annual inventory cost is minimized, the annual inventory holding cost (in Rs.) is _____

Correct Answer :

34000 to 36000

Question Number : 35 Question Type : MCQ

Sales data of a product is given in the following table:

Month	January	February	March	April	May
Number of units sold	10	11	16	19	25

Regarding forecast for the month of June, which one of the following statements is **TRUE**?

(A) Moving average will forecast a higher value compared to regression.
(B) Higher the value of order N, the greater will be the forecast value by moving average.
(C) Exponential smoothing will forecast a higher value compared to regression.
(D) Regression will forecast a higher value compared to moving average.

1. ✖ A
2. ✖ B
3. ✖ C
4. ✔ D

Question Number : 36 Question Type : MCQ

The chance of a student passing an exam is 20%. The chance of a student passing the exam and getting above 90% marks in it is 5%. GIVEN that a student passes the examination, the probability that the student gets above 90% marks is

(A) $\frac{1}{18}$ (B) $\frac{1}{4}$ (C) $\frac{2}{9}$ (D) $\frac{5}{18}$

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 37 Question Type : NAT

The surface integral $\iint_S \frac{1}{\pi}(9x\mathbf{i} - 3y\mathbf{j}) \bullet \mathbf{n}\, dS$ over the sphere given by $x^2 + y^2 + z^2 = 9$ is _____

Correct Answer :

214 to 218

Question Number : 38 Question Type : MCQ

Consider the following differential equation:

$$\frac{dy}{dt} = -5y; \quad \text{initial condition: } y = 2 \text{ at } t = 0.$$

The value of y at $t = 3$ is

(A) $-5e^{-10}$ (B) $2e^{-10}$ (C) $2e^{-15}$ (D) $-15e^{2}$

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 39 Question Type : NAT

The values of function $f(x)$ at 5 discrete points are given below:

x	0	0.1	0.2	0.3	0.4
$f(x)$	0	10	40	90	160

Using Trapezoidal rule with step size of 0.1, the value of $\int_0^{0.4} f(x)\,dx$ is _____

Correct Answer :

21.8 to 22.2

Question Number : 40 Question Type : NAT

The initial velocity of an object is 40 m/s. The acceleration a of the object is given by the following expression:

$$a = -0.1v,$$

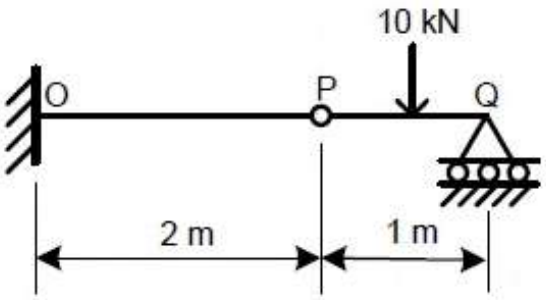
where v is the instantaneous velocity of the object. The velocity of the object after 3 seconds will be _____

Correct Answer :

29.5 to 29.7

Question Number : 41 Question Type : MCQ

A cantilever beam OP is connected to another beam PQ with a pin joint as shown in the figure. A load of 10 kN is applied at the mid-point of PQ. The magnitude of bending moment (in kN-m) at fixed end O is



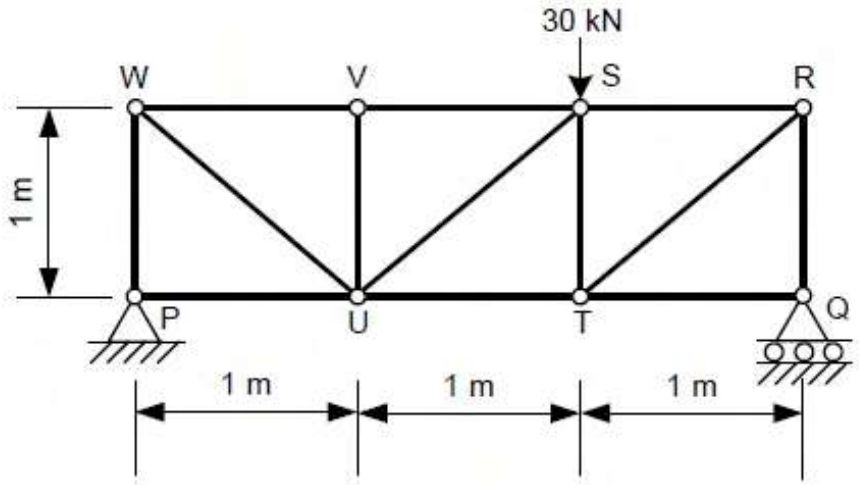
(A) 2.5 (B) 5 (C) 10 (D) 25

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 42 Question Type : MCQ

For the truss shown in the figure, the magnitude of the force (in kN) in the member SR is



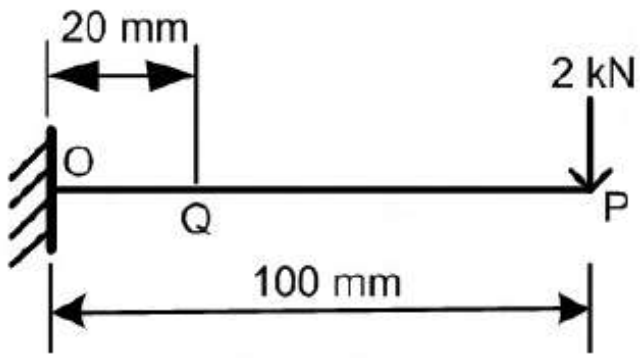
(A) 10 (B) 14.14 (C) 20 (D) 28.28

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 43 Question Type : NAT

A cantilever beam with square cross-section of 6 mm side is subjected to a load of 2 kN normal to the top surface as shown in the figure. The Young's modulus of elasticity of the material of the beam is 210 GPa. The magnitude of slope (in radian) at Q (20 mm from the fixed end) is _____



Correct Answer :
0.15 to 0.17

Question Number : 44 Question Type : MCQ

In a plane stress condition, the components of stress at a point are $\sigma_x = 20$ MPa, $\sigma_y = 80$ MPa and $\tau_{xy} = 40$ MPa. The maximum shear stress (in MPa) at the point is

(A) 20 (B) 25 (C) 50 (D) 100

Options :
1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 45 Question Type : NAT

In a certain slider-crank mechanism, lengths of crank and connecting rod are equal. If the crank rotates with a uniform angular speed of 14 rad/s and the crank length is 300 mm, the maximum acceleration of the slider (in m/s^2) is _____

Correct Answer :115 to 120

Question Number : 46 Question Type : MCQ

A single-degree-freedom spring-mass system is subjected to a sinusoidal force of 10 N amplitude and frequency ω along the axis of the spring. The stiffness of the spring is 150 N/m, damping factor is 0.2 and the undamped natural frequency is 10ω. At steady state, the amplitude of vibration (in m) is approximately

(A) 0.05 (B) 0.07 (C) 0.70 (D) 0.90

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 47 Question Type : NAT

A hollow shaft of 1 m length is designed to transmit a power of 30 kW at 700 rpm. The maximum permissible angle of twist in the shaft is 1°. The inner diameter of the shaft is 0.7 times the outer diameter. The modulus of rigidity is 80 GPa. The outside diameter (in mm) of the shaft is_____

Correct Answer :

43 to 45

Question Number : 48 Question Type : MCQ

A hollow shaft ($d_o = 2d_i$ where d_o and d_i are the outer and inner diameters respectively) needs to transmit 20 kW power at 3000 RPM. If the maximum permissible shear stress is 30 MPa, d_o is

(A) 11.29 mm (B) 22.58 mm (C) 33.87 mm (D) 45.16 mm

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 49 Question Type : MCQ

The total emissive power of a surface is 500 W/m^2 at a temperature T_1 and 1200 W/m^2 at a temperature T_2, where the temperatures are in Kelvin. Assuming the emissivity of the surface to be constant, the ratio of the temperatures $\frac{T_1}{T_2}$ is

(A) 0.308 (B) 0.416 (C) 0.803 (D) 0.874

Options :

1. ✖ A
2. ✖ B
3. ✔ C

4. ✖ D

Question Number : 50 Question Type : MCQ

The head loss for a laminar incompressible flow through a horizontal circular pipe is h_1. Pipe length and fluid remaining the same, if the average flow velocity doubles and the pipe diameter reduces to half its previous value, the head loss is h_2. The ratio h_2/h_1 is

(A) 1 (B) 4 (C) 8 (D) 16

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 51 Question Type : NAT

For a fully developed laminar flow of water (dynamic viscosity 0.001 Pa-s) through a pipe of radius 5 cm, the axial pressure gradient is −10 Pa/m. The magnitude of axial velocity (in m/s) at a radial location of 0.2 cm is _____

Correct Answer :

6.2 to 6.3

Question Number : 52 Question Type : MCQ

A balanced counterflow heat exchanger has a surface area of 20 m^2 and overall heat transfer coefficient of 20 W/m^2-K. Air (C_p=1000 J/kg-K) entering at 0.4 kg/s and 280 K is to be preheated by the air leaving the system at 0.4 kg/s and 300 K. The outlet temperature (in K) of the preheated air is

(A) 290 (B) 300 (C) 320 (D) 350

Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 53 Question Type : MCQ

A cylindrical uranium fuel rod of radius 5 mm in a nuclear reactor is generating heat at the rate of 4×10^7 W/m^3. The rod is cooled by a liquid (convective heat transfer coefficient 1000 W/m^2-K) at 25 °C. At steady state, the surface temperature (in K) of the rod is

(A) 308 (B) 398 (C) 418 (D) 448

Options :

1. ✖ A

2. ✔ B

3. ✖ C

4. ✖ D

Question Number : 54 Question Type : NAT

Work is done on an adiabatic system due to which its velocity changes from 10 m/s to 20 m/s, elevation increases by 20 m and temperature increases by 1 K. The mass of the system is 10 kg, C_v = 100 J/(kg.K) and gravitational acceleration is 10 m/s^2. If there is no change in any other component of the energy of the system, the magnitude of total work done (in kJ) on the system is _____

Correct Answer :

4.5

Question Number : 55 Question Type : NAT

One kg of air (R = 287 J/kg-K) undergoes an irreversible process between equilibrium state 1 (20 °C, 0.9 m^3) and equilibrium state 2 (20 °C, 0.6 m^3). The change in entropy $s_2 - s_1$ (in J/kg-K) is _____

Correct Answer :

-117 to -115

Question Number : 56 Question Type : MCQ

For the same values of peak pressure, peak temperature and heat rejection, the correct order of efficiencies for Otto, Dual and Diesel cycles is

(A) $\eta_{Otto} > \eta_{Dual} > \eta_{Diesel}$

(B) $\eta_{Diesel} > \eta_{Dual} > \eta_{Otto}$

(C) $\eta_{Dual} > \eta_{Diesel} > \eta_{Otto}$

(D) $\eta_{Diesel} > \eta_{Otto} > \eta_{Dual}$

Options :

1. ✖ A

2. ✔ B

3. ✖ C

4. ✖ D

Question Number : 57 Question Type : NAT

In a Rankine cycle, the enthalpies at turbine entry and outlet are 3159 kJ/kg and 2187 kJ/kg, respectively. If the specific pump work is 2 kJ/kg, the specific steam consumption (in kg/kW-h) of the cycle based on net output is _____

Correct Answer :

3.6 to 3.8

Question Number : 58 Question Type : NAT

A cube and a sphere made of cast iron (each of volume 1000 cm^3) were cast under identical conditions. The time taken for solidifying the cube was 4 s. The solidification time (in s) for the sphere is _____

Correct Answer :

6.0 to 6.3

Question Number : 59 Question Type : MCQ

In a two-stage wire drawing operation, the fractional reduction (ratio of change in cross-sectional area to initial cross-sectional area) in the first stage is 0.4. The fractional reduction in the second stage is 0.3. The overall fractional reduction is

(A) 0.24 (B) 0.58 (C) 0.60 (D) 1.00

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 60 Question Type : NAT

The flow stress (in MPa) of a material is given by

$$\sigma = 500\,\varepsilon^{0.1},$$

where ε is true strain. The Young's modulus of elasticity of the material is 200 GPa. A block of thickness 100 mm made of this material is compressed to 95 mm thickness and then the load is removed. The final dimension of the block (in mm) is _____

Correct Answer :

95.14 to 95.20

Question Number : 61 Question Type : MCQ

During a TIG welding process, the arc current and arc voltage were 50 A and 60 V, respectively, when the welding speed was 150 mm/min. In another process, the TIG welding is carried out at a welding speed of 120 mm/min at the same arc voltage and heat input to the material so that weld quality remains the same. The welding current (in A) for this process is

(A) 40.00 (B) 44.72 (C) 55.90 (D) 62.25

Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 62 Question Type : NAT

A single point cutting tool with 0° rake angle is used in an orthogonal machining process. At a cutting speed of 180 m/min, the thrust force is 490 N. If the coefficient of friction between the tool and the chip is 0.7, then the power consumption (in kW) for the machining operation is _____

Correct Answer :

2.0 to 2.2

Question Number : 63 Question Type : MCQ

A resistance-capacitance relaxation circuit is used in an electrical discharge machining process. The discharge voltage is 100 V. At a spark cycle time of 25 μs, the average power input required is 1 kW. The capacitance (in μF) in the circuit is

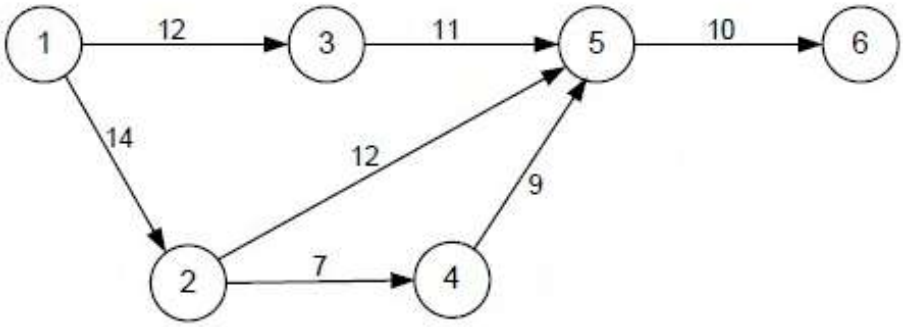
(A) 2.5 (B) 5.0 (C) 7.5 (D) 10.0

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 64 Question Type : NAT

A project consists of 7 activities. The network along with the time durations (in days) for various activities is shown in the figure.



The minimum time (in days) for completion of the project is_____

Correct Answer :

39 to 40

Question Number : 65 Question Type : NAT

A manufacturer has the following data regarding a product:
Fixed cost per month = Rs. 50000
Variable cost per unit = Rs. 200
Selling price per unit = Rs. 300
Production capacity = 1500 units per month
If the production is carried out at 80% of the rated capacity, then the monthly profit (in Rs.) is _____

Correct Answer :

68000 to 72000

SESSION - 3

Graduate Aptitude Test in Engineering

Notations :

1.Options shown in green color and with ✔ icon are correct.

2.Options shown in red color and with ✖ icon are incorrect.

Question Paper Name: ME: MECHANICAL ENGINEERING 1st Feb shift2
Number of Questions: 65
Total Marks: 100.0

Wrong answer for MCQ will result in negative marks, (-1/3) for 1 mark Questions and (-2/3) for 2 marks Questions.

General Aptitude

Number of Questions: 10
Section Marks: 15.0

Q.1 to Q.5 carry 1 mark each & Q.6 to Q.10 carry 2 marks each.

Question Number : 1 Question Type : MCQ

Choose the appropriate word/phrase, out of the four options given below, to complete the following sentence:

Apparent lifelessness _____ dormant life.

(A) harbours (B) leads to (C) supports (D) affects

Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 2 Question Type : MCQ

Fill in the blank with the correct idiom/phrase.

That boy from the town was a _____ in the sleepy village.

(A) dog out of herd (B) sheep from the heap
(C) fish out of water (D) bird from the flock

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 3 Question Type : MCQ

Choose the statement where underlined word is used correctly.

(A) When the teacher eludes to different authors, he is being elusive.
(B) When the thief keeps eluding the police, he is being elusive.
(C) Matters that are difficult to understand, identify or remember are allusive.
(D) Mirages can be allusive, but a better way to express them is illusory.

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 4 Question Type : MCQ

Tanya is older than Eric.
Cliff is older than Tanya.
Eric is older than Cliff.

If the first two statements are true, then the third statement is:

(A) True
(B) False
(C) Uncertain
(D) Data insufficient

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 5 Question Type : MCQ

Five teams have to compete in a league, with every team playing every other team exactly once, before going to the next round. How many matches will have to be held to complete the league round of matches?

(A) 20 (B) 10 (C) 8 (D) 5

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 6 Question Type : MCQ

Select the appropriate option in place of underlined part of the sentence.

Increased productivity necessary reflects greater efforts made by the employees.

(A) Increase in productivity necessary
(B) Increase productivity is necessary
(C) Increase in productivity necessarily
(D) No improvement required

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 7 Question Type : MCQ

Given below are two statements followed by two conclusions. Assuming these statements to be true, decide which one logically follows.

Statements:

I. No manager is a leader.
II. All leaders are executives.

Conclusions:

I. No manager is an executive.
II. No executive is a manager.

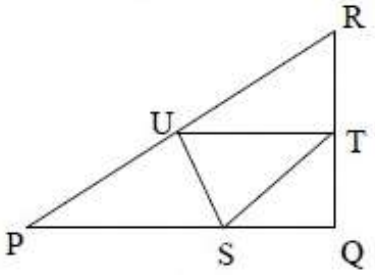
(A) Only conclusion I follows.
(B) Only conclusion II follows.
(C) Neither conclusion I nor II follows.
(D) Both conclusions I and II follow.

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 8 Question Type : NAT

In the given figure angle Q is a right angle, PS:QS = 3:1, RT:QT = 5:2 and PU:UR = 1:1. If area of triangle QTS is 20 cm^2, then the area of triangle PQR in cm^2 is _____.



Correct Answer :

280

Question Number : 9 Question Type : MCQ

Right triangle PQR is to be constructed in the xy - plane so that the right angle is at P and line PR is parallel to the x-axis. The x and y coordinates of P, Q, and R are to be integers that satisfy the inequalities: $-4 \le x \le 5$ and $6 \le y \le 16$. How many different triangles could be constructed with these properties?

(A) 110 (B) 1,100 (C) 9,900 (D) 10,000

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 10 Question Type : MCQ

A coin is tossed thrice. Let X be the event that head occurs in each of the first two tosses. Let Y be the event that a tail occurs on the third toss. Let Z be the event that two tails occur in three tosses. Based on the above information, which one of the following statements is TRUE?

(A) X and Y are not independent
(B) Y and Z are dependent
(C) Y and Z are independent
(D) X and Z are independent

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

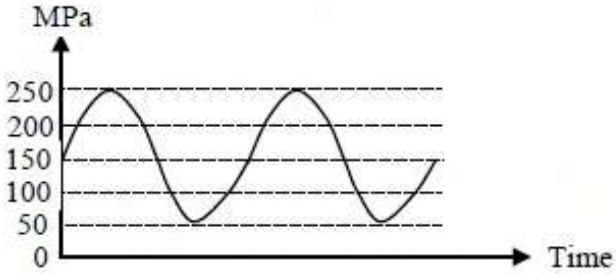
Mechanical Engineering

Number of Questions:	55
Section Marks:	85.0

Q.11 to Q.35 carry 1 mark each & Q.36 to Q.65 carry 2 marks each.

Question Number : 11 Question Type : MCQ

For the given fluctuating fatigue load, the values of stress amplitude and stress ratio are respectively



(A) 100 MPa and 5
(B) 250 MPa and 5
(C) 100 MPa and 0.20
(D) 250 MPa and 0.20

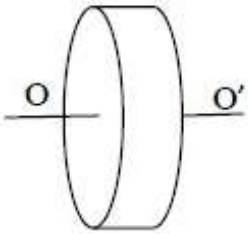
Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

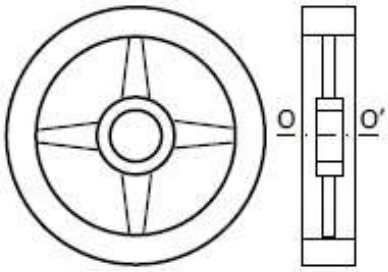
Question Number : 12 Question Type : MCQ

For the same material and the mass, which of the following configurations of flywheel will have maximum mass moment of inertia about the axis of rotation OO′ passing through the center of gravity.

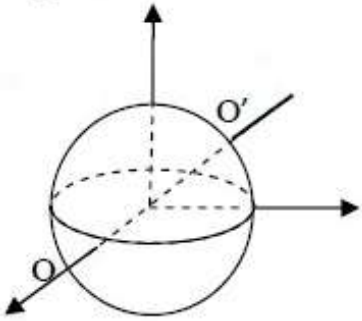
(A) Solid Cylinder



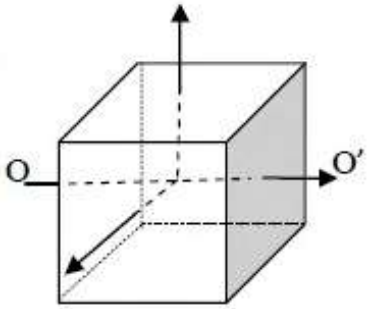
(B) Rimmed wheel



(C) Solid sphere



(D) Solid cube

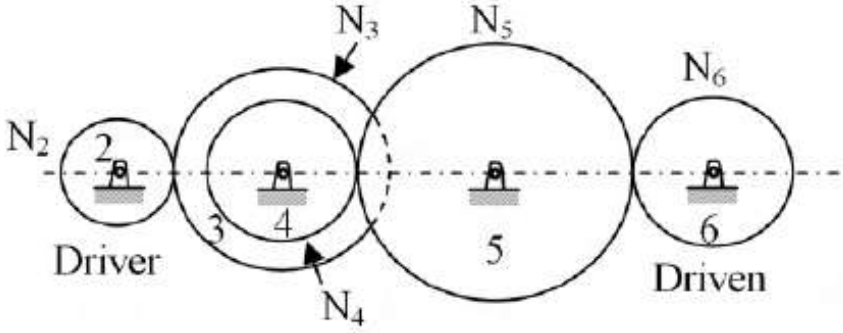


Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 13 Question Type : MCQ

A gear train is made up of five spur gears as shown in the figure. Gear 2 is driver and gear 6 is driven member. N_2, N_3, N_4, N_5 and N_6 represent number of teeth on gears 2, 3, 4, 5, and 6 respectively. The gear(s) which act(s) as idler(s) is/are



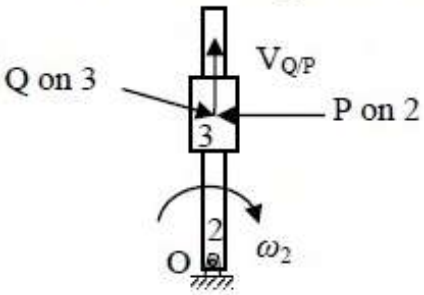
(A) Only 3 (B) Only 4 (C) Only 5 (D) Both 3 and 5

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 14 Question Type : MCQ

In the figure, link 2 rotates with constant angular velocity ω_2. A slider link 3 moves outwards with a constant relative velocity $V_{Q/P}$, where Q is a point on slider 3 and P is a point on link 2. The magnitude and direction of Coriolis component of acceleration is given by



(A) $2\,\omega_2\,V_{Q/P}$; direction of $V_{Q/P}$ rotated by 90° in the direction of ω_2

(B) $\omega_2\,V_{Q/P}$; direction of $V_{Q/P}$ rotated by 90° in the direction of ω_2

(C) $2\,\omega_2\,V_{Q/P}$; direction of $V_{Q/P}$ rotated by 90° opposite to the direction of ω_2

(D) $\omega_2\,V_{Q/P}$; direction of $V_{Q/P}$ rotated by 90° opposite to the direction of ω_2

Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 15 Question Type : MCQ

The strain hardening exponent n of stainless steel SS 304 with distinct yield and UTS values undergoing plastic deformation is

(A) $n < 0$ (B) $n = 0$ (C) $0 < n < 1$ (D) $n = 1$

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 16 Question Type : MCQ

In a machining operation, if the generatrix and directrix both are straight lines, the surface obtained is

(A) cylindrical (B) helical (C) plane (D) surface of revolution

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 17 Question Type : MCQ

In full mould (cavity-less) casting process, the pattern is made of

(A) expanded polystyrene (B) wax
(C) epoxy (D) plaster of Paris

Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 18 Question Type : MCQ

In the notation (a/b/c) : (d/e/f) for summarizing the characteristics of queueing situation, the letters 'b' and 'd' stand respectively for

(A) service time distribution and queue discipline
(B) number of servers and size of calling source
(C) number of servers and queue discipline
(D) service time distribution and maximum number allowed in system

Options :

1. ✔ A
2. ✖ B

3. ✖ C
4. ✖ D

Question Number : 19 Question Type : MCQ

Couette flow is characterized by

(A) steady, incompressible, laminar flow through a straight circular pipe

(B) fully developed turbulent flow through a straight circular pipe

(C) steady, incompressible, laminar flow between two fixed parallel plates

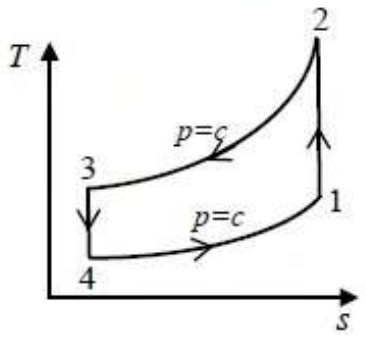
(D) steady, incompressible, laminar flow between one fixed plate and the other moving with a constant velocity

Options :

1. ✖ A
2. ✖ B
3. ✖ C
4. ✔ D

Question Number : 20 Question Type : MCQ

The thermodynamic cycle shown in figure (T-s diagram) indicates



(A) reversed Carnot cycle
(B) reversed Brayton cycle
(C) vapor compression cycle
(D) vapor absorption cycle

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 21 Question Type : MCQ

The ratio of momentum diffusivity (ν) to thermal diffusivity (α), is called

(A) Prandtl number
(B) Nusselt number
(C) Biot number
(D) Lewis number

Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 22 Question Type : MCQ

Saturated vapor is condensed to saturated liquid in a condenser. The heat capacity ratio is $C_r = \frac{C_{min}}{C_{max}}$. The effectiveness (ε) of the condenser is

(A) $\frac{1-\exp[-NTU(1+C_r)]}{1+C_r}$

(B) $\frac{1-\exp[-NTU(1-C_r)]}{1-C_r\exp[-NTU(1-C_r)]}$

(C) $\frac{NTU}{1+NTU}$

(D) $1-\exp(-NTU)$

Options :

1. ✖ A
2. ✖ B
3. ✖ C
4. ✔ D

Question Number : 23 Question Type : NAT

Using a unit step size, the value of integral $\int_1^2 x \ln x \, dx$ by trapezoidal rule is _____

Correct Answer :

0.68 to 0.70

Question Number : 24 Question Type : MCQ

If P(X) = 1/4, P(Y) = 1/3, and P(X ∩ Y) = 1/12, the value of P(Y/X) is

(A) $\frac{1}{4}$

(B) $\frac{4}{25}$

(C) $\frac{1}{3}$

(D) $\frac{29}{50}$

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 25 Question Type : NAT

The lowest eigenvalue of the 2×2 matrix $\begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix}$ is _____

Correct Answer:
2

Question Number : 26 Question Type : NAT

The value of $\lim_{x \to 0} \left(\frac{-\sin x}{2 \sin x + x \cos x} \right)$ is _____

Correct Answer :
-0.35 to -0.30

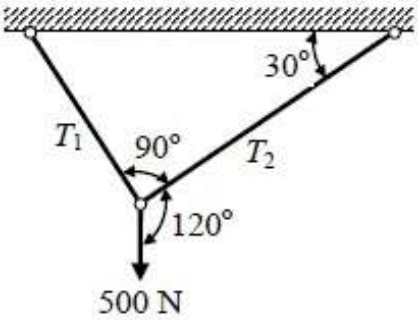
Question Number : 27 Question Type : NAT

A cylindrical tank with closed ends is filled with compressed air at a pressure of 500 kPa. The inner radius of the tank is 2 m, and it has wall thickness of 10 mm. The magnitude of maximum in-plane shear stress (in MPa) is _____

Correct Answer :
25

Question Number : 28 Question Type : MCQ

A weight of 500 N is supported by two metallic ropes as shown in the figure. The values of tensions T_1 and T_2 are respectively



(A) 433 N and 250 N
(B) 250 N and 433 N
(C) 353.5 N and 250 N
(D) 250 N and 353.5 N

Options :
1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 29 Question Type : MCQ

Which of the following statements are **TRUE** for damped vibrations?

P. For a system having critical damping, the value of damping ratio is unity and system does not undergo a vibratory motion.

Q. Logarithmic decrement method is used to determine the amount of damping in a physical system.

R. In case of damping due to dry friction between moving surfaces resisting force of constant magnitude acts opposite to the relative motion.

S. For the case of viscous damping, drag force is directly proportional to the square of relative velocity.

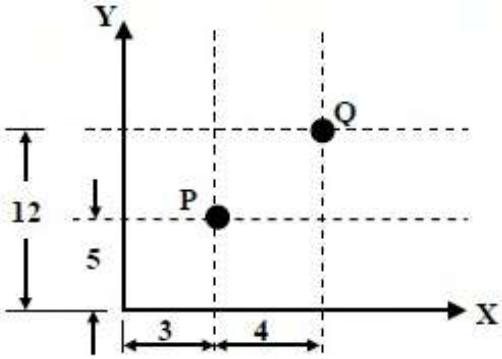
(A) P and Q only
(B) P and S only
(C) P, Q and R only
(D) Q and S only

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 30 Question Type : MCQ

A drill is positioned at point P and it has to proceed to point Q. The coordinates of point Q in the incremental system of defining position of a point in CNC part program will be



(A) (3, 12)
(B) (5, 7)
(C) (7, 12)
(D) (4, 7)

Options :

1. ✖ A
2. ✖ B
3. ✖ C
4. ✔ D

Question Number : 31 Question Type : MCQ

Which two of the following joining processes are autogeneous?

(i) Diffusion welding
(ii) Electroslag welding
(iii) Tungsten inert gas welding
(iv) Friction welding

(A) (i) and (iv) (B) (ii) and (iii) (C) (ii) and (iv) (D) (i) and (iii)

Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 32 Question Type : MCQ

Three parallel pipes connected at the two ends have flow-rates Q_1, Q_2 and Q_3 respectively, and the corresponding frictional head losses are h_{L1}, h_{L2} and h_{L3} respectively. The correct expressions for total flow rate (Q) and frictional head loss across the two ends (h_L) are

(A) $Q = Q_1 + Q_2 + Q_3$; $h_L = h_{L1} + h_{L2} + h_{L3}$
(B) $Q = Q_1 + Q_2 + Q_3$; $h_L = h_{L1} = h_{L2} = h_{L3}$
(C) $Q = Q_1 = Q_2 = Q_3$; $h_L = h_{L1} + h_{L2} + h_{L3}$
(D) $Q = Q_1 = Q_2 = Q_3$; $h_L = h_{L1} = h_{L2} = h_{L3}$

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 33 Question Type : MCQ

A rigid container of volume 0.5 m^3 contains 1.0 kg of water at 120°C (v_f = 0.00106 m^3/kg, v_g= 0.8908 m^3/kg). The state of water is

(A) compressed liquid
(B) saturated liquid
(C) a mixture of saturated liquid and saturated vapor
(D) superheated vapor

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 34 Question Type : MCQ

Let ϕ be an arbitrary smooth real valued scalar function and $\vec{V}$ be an arbitrary smooth vector valued function in a three-dimensional space. Which one of the following is an identity?

(A) $\text{Curl}\left(\phi\vec{V}\right) = \nabla\left(\phi\ \text{Div}\vec{V}\right)$

(B) $\text{Div}\vec{V} = 0$

(C) $\text{Div Curl}\vec{V} = 0$

(D) $\text{Div}\left(\phi\vec{V}\right) = \phi\ \text{Div}\vec{V}$

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 35 Question Type : MCQ

An air-standard Diesel cycle consists of the following processes:
1-2: Air is compressed isentropically.
2-3: Heat is added at constant pressure.
3-4: Air expands isentropically to the original volume.
4-1: Heat is rejected at constant volume.
If γ and T denote the specific heat ratio and temperature, respectively, the efficiency of the cycle is

(A) $1 - \dfrac{T_4 - T_1}{T_3 - T_2}$

(B) $1 - \dfrac{T_4 - T_1}{\gamma\left(T_3 - T_2\right)}$

(C) $1 - \dfrac{\gamma\left(T_4 - T_1\right)}{T_3 - T_2}$

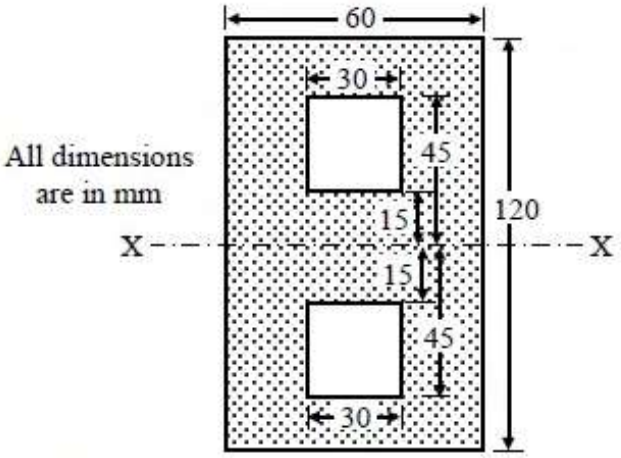
(D) $1 - \dfrac{T_4 - T_1}{\left(\gamma - 1\right)\left(T_3 - T_2\right)}$

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 36 Question Type : MCQ

The value of moment of inertia of the section shown in the figure about the axis-XX is



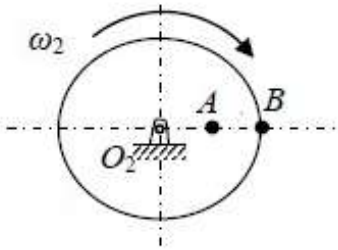
(A) 8.5050×10^6 mm^4
(B) 6.8850×10^6 mm^4
(C) 7.7625×10^6 mm^4
(D) 8.5725×10^6 mm^4

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 37 Question Type : NAT

Figure shows a wheel rotating about O_2. Two points A and B located along the radius of wheel have speeds of 80 m/s and 140 m/s respectively. The distance between the points A and B is 300 mm. The diameter of the wheel (in mm) is _____

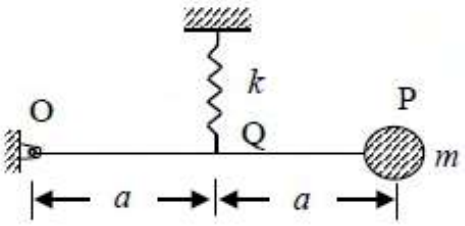


Correct Answer :

1390 to 1410

Question Number : 38 Question Type : MCQ

Figure shows a single degree of freedom system. The system consists of a massless rigid bar OP hinged at O and a mass m at end P. The natural frequency of vibration of the system is



(A) $f_n = \frac{1}{2\pi}\sqrt{\frac{k}{4m}}$

(B) $f_n = \frac{1}{2\pi}\sqrt{\frac{k}{2m}}$

(C) $f_n = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$

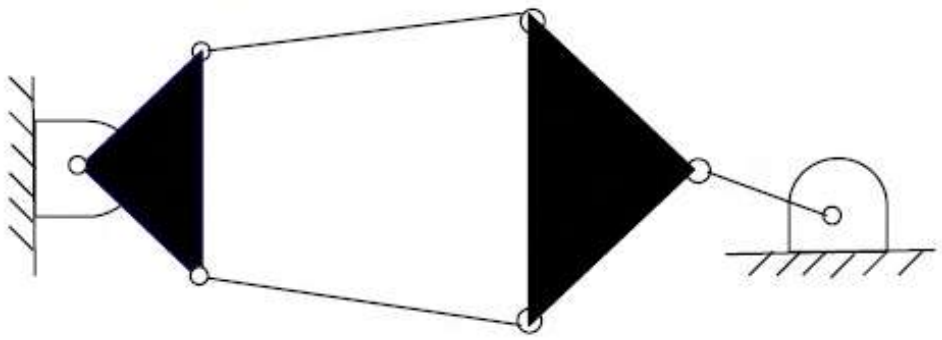
(D) $f_n = \frac{1}{2\pi}\sqrt{\frac{2k}{m}}$

Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 39 Question Type : MCQ

The number of degrees of freedom of the linkage shown in the figure is



(A) −3 (B) 0 (C) 1 (D) 2

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 40 Question Type : NAT

For ball bearings, the fatigue life L measured in number of revolutions and the radial load F are related by $FL^{1/3} = K$, where K is a constant. It withstands a radial load of 2 kN for a life of 540 million revolutions. The load (in kN) for a life of one million revolutions is _____

Correct Answer:
15 to 17

Question Number : 41 Question Type : NAT

In a rolling operation using rolls of diameter 500 mm, if a 25 mm thick plate cannot be reduced to less than 20 mm in one pass, the coefficient of friction between the roll and the plate is _____

Correct Answer :
0.10 to 0.15

Question Number : 42 Question Type : NAT

Ratio of solidification time of a cylindrical casting (height = radius) to that of a cubic casting of side two times the height of cylindrical casting is _____

Correct Answer :
0.5 to 0.6

Question Number : 43 Question Type : NAT

The annual requirement of rivets at a ship manufacturing company is 2000 kg. The rivets are supplied in units of 1 kg costing Rs. 25 each. If it costs Rs. 100 to place an order and the annual cost of carrying one unit is 9% of its purchase cost, the cycle length of the order (in days) will be _____

Correct Answer :
76 to 78

Question Number : 44 Question Type : MCQ

Orthogonal turning of a mild steel tube with a tool of rake angle 10° is carried out at a feed of 0.14 mm/rev. If the thickness of the chip produced is 0.28 mm, the values of shear angle and shear strain will be respectively

(A) 28°20′ and 2.19 (B) 22°20′ and 3.53 (C) 24°30′ and 4.19 (D) 37°20′ and 5.19

Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 45 Question Type : MCQ

In a CNC milling operation, the tool has to machine the circular arc from point (20, 20) to (10, 10) at sequence number 5 of the CNC part program. If the center of the arc is at (20, 10) and the machine has incremental mode of defining position coordinates, the correct tool path command is

(A) N 05 G90 G01 X–10 Y–10 R10

(B) N 05 G91 G03 X–10 Y–10 R10

(C) N 05 G90 G03 X20 Y20 R10

(D) N 05 G91 G02 X20 Y20 R10

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 46 Question Type : NAT

A Prandtl tube (Pitot-static tube with C=1) is used to measure the velocity of water. The differential manometer reading is 10 mm of liquid column with a relative density of 10. Assuming $g = 9.8$ m/s^2, the velocity of water (in m/s) is _____

Correct Answer :

1.30 to 1.34

Question Number : 47 Question Type : NAT

Refrigerant vapor enters into the compressor of a standard vapor compression cycle at − 10°C (h = 402 kJ/kg) and leaves the compressor at 50°C (h = 432 kJ/kg). It leaves the condenser at 30°C (h = 237 kJ/kg). The COP of the cycle is _____

Correct Answer:
5.5

Question Number : 48 Question Type : NAT

Steam enters a turbine at 30 bar, 300°C (u = 2750 kJ/kg, h = 2993 kJ/kg) and exits the turbine as saturated liquid at 15 kPa (u = 225 kJ/kg, h = 226 kJ/kg). Heat loss to the surrounding is 50 kJ/kg of steam flowing through the turbine. Neglecting changes in kinetic energy and potential energy, the work output of the turbine (in kJ/kg of steam) is _____

Correct Answer :
2717

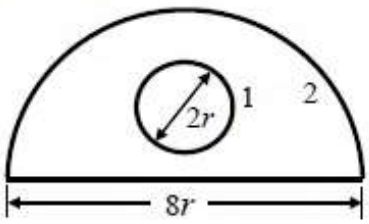
Question Number : 49 Question Type : NAT

Air in a room is at 35°C and 60% relative humidity (RH). The pressure in the room is 0.1 MPa. The saturation pressure of water at 35°C is 5.63 kPa. The humidity ratio of the air (in gram/kg of dry air) is _____

Correct Answer :
21.7 to 21.9

Question Number : 50 Question Type : MCQ

A solid sphere 1 of radius 'r' is placed inside a hollow, closed hemispherical surface 2 of radius '$4r$'. The shape factor $F_{2\text{-}1}$ is



(A) 1/12 (B) 1/2 (C) 2 (D) 12

Options :
1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 51 Question Type : NAT

The value of

$\int_C [(3x - 8y^2)dx + (4y - 6xy)dy]$, (where C is the boundary of the region bounded by $x = 0$, $y = 0$ and $x+y = 1$) is _____

Correct Answer :

1.60 to 1.70

Question Number : 52 Question Type : MCQ

For a given matrix $P = \begin{bmatrix} 4+3i & -i \\ i & 4-3i \end{bmatrix}$, where $i = \sqrt{-1}$, the inverse of matrix P is

(A) $\frac{1}{24}\begin{bmatrix} 4-3i & i \\ -i & 4+3i \end{bmatrix}$

(B) $\frac{1}{25}\begin{bmatrix} i & 4-3i \\ 4+3i & -i \end{bmatrix}$

(C) $\frac{1}{24}\begin{bmatrix} 4+3i & -i \\ i & 4-3i \end{bmatrix}$

(D) $\frac{1}{25}\begin{bmatrix} 4+3i & -i \\ i & 4-3i \end{bmatrix}$

Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 53 Question Type : NAT

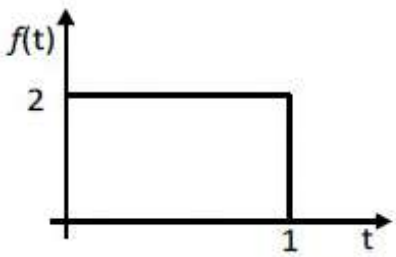
Newton-Raphson method is used to find the roots of the equation, $x^3 + 2x^2 + 3x - 1 = 0$. If the initial guess is $x_0 = 1$, then the value of x after 2nd iteration is _____

Correct Answer :

0.29 to 0.31

Question Number : 54 Question Type : MCQ

Laplace transform of the function $f(t)$ is given by $F(s) = L\{f(t)\} = \int_0^\infty f(t)e^{-st}\,dt$.
Laplace transform of the function shown below is given by



(A) $\frac{1-e^{-2s}}{s}$

(B) $\frac{1-e^{-s}}{2s}$

(C) $\frac{2-2e^{-s}}{s}$

(D) $\frac{1-2e^{-s}}{s}$

Options :

1. ✖ A

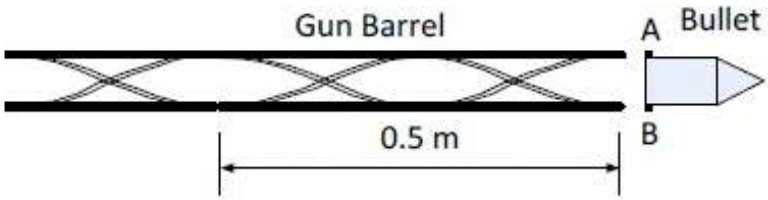
2. ✖ B

3. ✔ C

4. ✖ D

Question Number : 55 Question Type : NAT

A bullet spins as the shot is fired from a gun. For this purpose, two helical slots as shown in the figure are cut in the barrel. Projections A and B on the bullet engage in each of the slots.



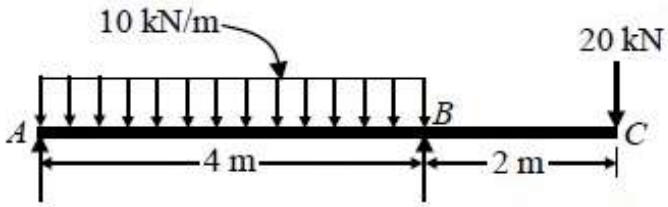
Helical slots are such that one turn of helix is completed over a distance of 0.5 m. If velocity of bullet when it exits the barrel is 20 m/s, its spinning speed in rad/s is _____

Correct Answer:

251 to 252

Question Number : 56 Question Type : NAT

For the overhanging beam shown in figure, the magnitude of maximum bending moment (in kN-m) is _____



Correct Answer:

40

Question Number : 57 Question Type : NAT

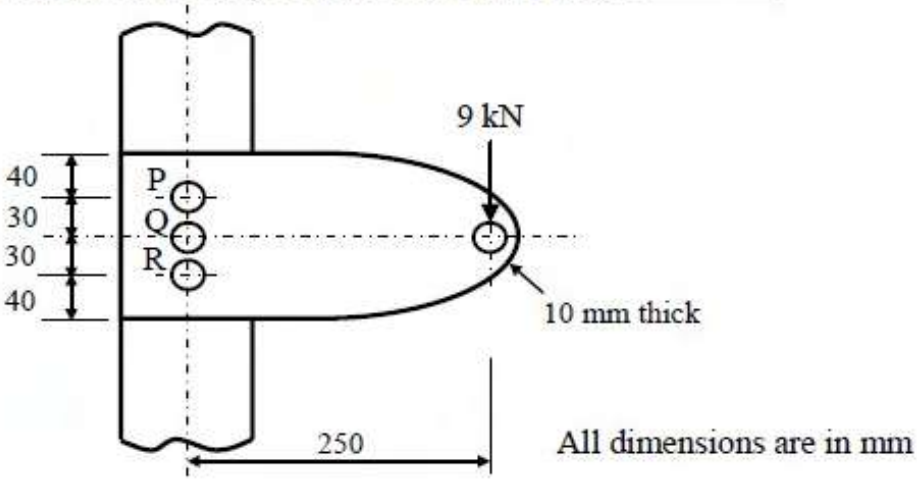
The torque (in N-m) exerted on the crank shaft of a two stroke engine can be described as $T = 10000 + 1000 \sin 2\theta - 1200 \cos 2\theta$, where θ is the crank angle as measured from inner dead center position. Assuming the resisting torque to be constant, the power (in kW) developed by the engine at 100 rpm is _____

Correct Answer:

104 to 105

Question Number : 58 Question Type : NAT

A cantilever bracket is bolted to a column using three M12×1.75 bolts P, Q and R. The value of maximum shear stress developed in the bolt P (in MPa) is _____



Correct Answer:

332 to 494

Question Number : 59 Question Type : MCQ

A shaft of length 90 mm has a tapered portion of length 55 mm. The diameter of the taper is 80 mm at one end and 65 mm at the other. If the taper is made by tailstock set over method, the taper angle and the set over respectively are

(A) 15°32′ and 12.16 mm
(B) 18°32′ and 15.66 mm
(C) 11°22′ and 10.26 mm
(D) 10°32′ and 14.46 mm

Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 60 Question Type : NAT

The dimensions of a cylindrical side riser (height = diameter) for a 25 cm × 15 cm × 5 cm steel casting are to be determined. For the tabulated shape factor values given below, the diameter of the riser (in cm) is _____

Shape factor	2	4	6	8	10	12
Riser volume/ Casting volume	1.0	0.70	0.55	0.50	0.40	0.35

Correct Answer:
10.5 to 10.7

Question Number : 61 Question Type : MCQ

For the linear programming problem:

$$\text{Maximize} \quad Z = 3X_1 + 2X_2$$

Subject to

$$-2X_1 + 3X_2 \leq 9$$
$$X_1 - 5X_2 \geq -20$$
$$X_1, X_2 \geq 0$$

The above problem has

(A) unbounded solution
(B) infeasible solution
(C) alternative optimum solution
(D) degenerate solution

Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 62 Question Type : MCQ

Which of the following statements are **TRUE**, when the cavitation parameter $\sigma = 0$?

(i) the local pressure is reduced to vapor pressure
(ii) cavitation starts
(iii) boiling of liquid starts
(iv) cavitation stops

(A) (i), (ii) and (iv)
(B) only (ii) and (iii)
(C) only (i) and (iii)
(D) (i), (ii) and (iii)

Options :

1. ✖ A
2. ✖ B
3. ✖ C
4. ✔ D

Question Number : 63 Question Type : NAT

One side of a wall is maintained at 400 K and the other at 300 K. The rate of heat transfer through the wall is 1000 W and the surrounding temperature is 25°C. Assuming no generation of heat within the wall, the irreversibility (in W) due to heat transfer through the wall is _____

Correct Answer:
247 to 249

Question Number : 64 Question Type : MCQ

A brick wall $\left(k = 0.9\frac{W}{m.K}\right)$ of thickness 0.18 m separates the warm air in a room from the cold ambient air. On a particular winter day, the outside air temperature is $-5°C$ and the room needs to be maintained at 27°C. The heat transfer coefficient associated with outside air is $20\frac{W}{m^2K}$. Neglecting the convective resistance of the air inside the room, the heat loss, in $\left(\frac{W}{m^2}\right)$, is

(A) 88 (B) 110 (C) 128 (D) 160

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 65 Question Type : NAT

A mixture of ideal gases has the following composition by mass:

N_2	O_2	CO_2
60%	30%	10%

If the universal gas constant is 8314 J/kmol-K, the characteristic gas constant of the mixture (in J/kg-K) is _____

Correct Answer :

274 to 276